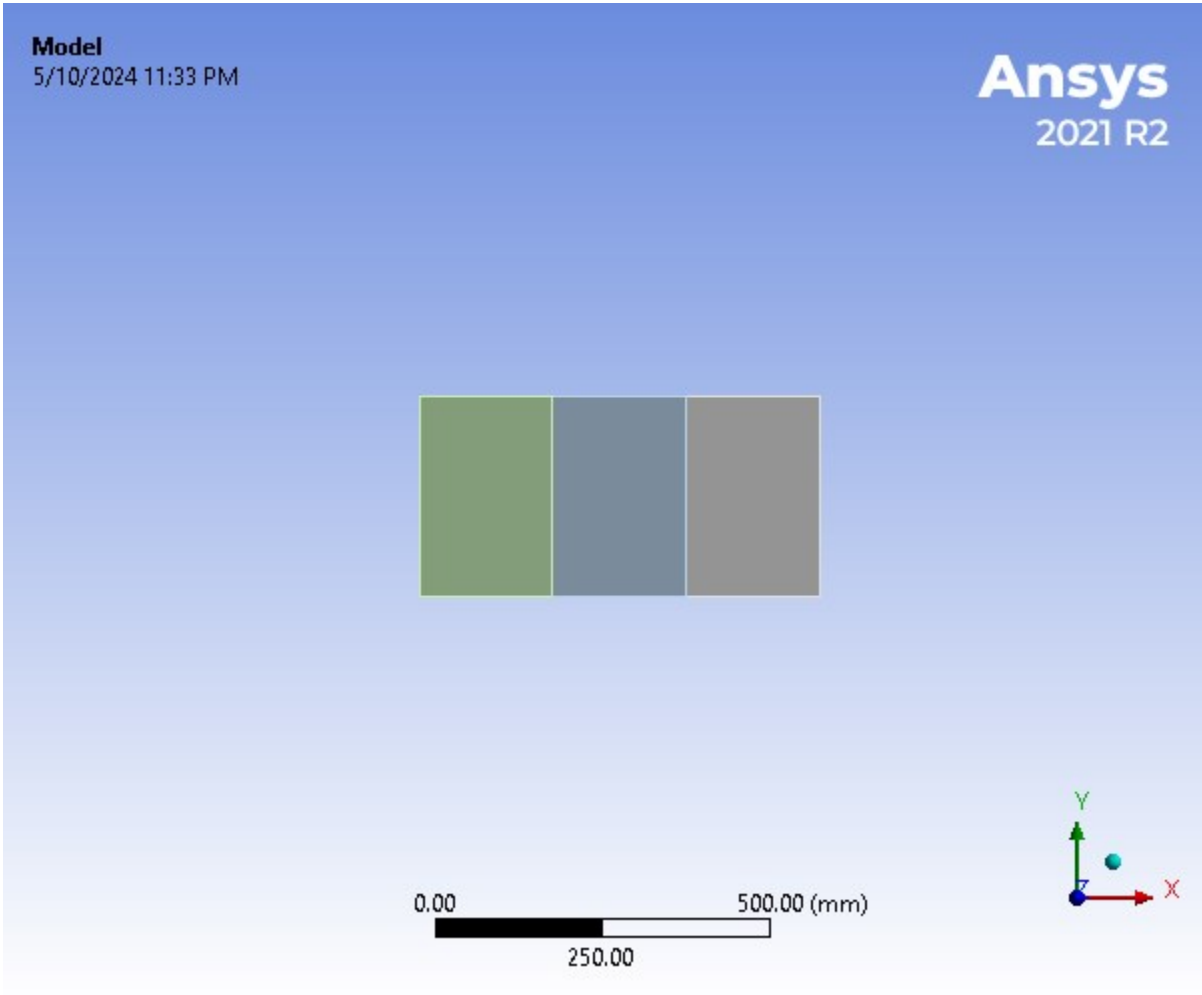




Project*

First Saved	Friday, May 10, 2024
Last Saved	Friday, May 10, 2024
Product Version	2021 R2
Save Project Before Solution	No
Save Project After Solution	No



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Units

TABLE 1

Unit System	Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4)

Geometry

TABLE 2
Model (A4) > Geometry

Object Name	<i>Geometry</i>
State	Fully Defined
Definition	
Source	C:\Users\90551\Desktop\5ler\18\2D_files\dp0\SYS\DM\SYS.agdb
Type	DesignModeler
Length Unit	Meters
Element Control	Program Controlled
2D Behavior	Plane Stress
Display Style	Body Color

Bounding Box	
Length X	600. mm
Length Y	300. mm
Properties	
Volume	1.8e+008 mm ³
Mass	1160.1 kg
Surface Area(approx.)	1.8e+005 mm ²
Scale Factor Value	1.
Statistics	
Bodies	3
Active Bodies	3
Nodes	735
Elements	210
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	2-D
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

TABLE 3
Model (A4) > Geometry > Parts

Model (1) - Geometry - Parts			
Object Name	2D	2D	2D
State	Meshed		
Graphics Properties			
Visible	Yes		
Transparency	1		
Definition			
Suppressed	No		
Stiffness Behavior	Flexible		

Coordinate System	Default Coordinate System		
Reference Temperature	By Environment		
Thickness	1000. mm		
Thickness Mode	Refresh on Update		
Treatment	None		
Material			
Assignment	Structural Steel	Aluminum Alloy	Bronze, C51000
Nonlinear Effects	Yes		
Thermal Strain Effects	Yes		
Bounding Box			
Length X	200. mm		
Length Y	300. mm		
Properties			
Volume	6.e+007 mm³		
Mass	471. kg	166.2 kg	522.9 kg
Centroid X	200. mm	-3.614e-015 mm	-200. mm
Centroid Y	0. mm		7.228e-015 mm
Centroid Z	0. mm		
Moment of Inertia Ip1	3.5325e+006 kg·mm²	1.2465e+006 kg·mm²	3.9218e+006 kg·mm²
Moment of Inertia Ip2	1.57e+006 kg·mm²	5.54e+005 kg·mm²	1.743e+006 kg·mm²
Moment of Inertia Ip3	5.1025e+006 kg·mm²	1.8005e+006 kg·mm²	5.6648e+006 kg·mm²
Surface Area(approx.)	60000 mm²		
Statistics			
Nodes	245		
Elements	70		
Mesh Metric	None		
CAD Attributes			
DMSheetThickness	0		

FIGURE 1
Model (A4) > Geometry > Figure

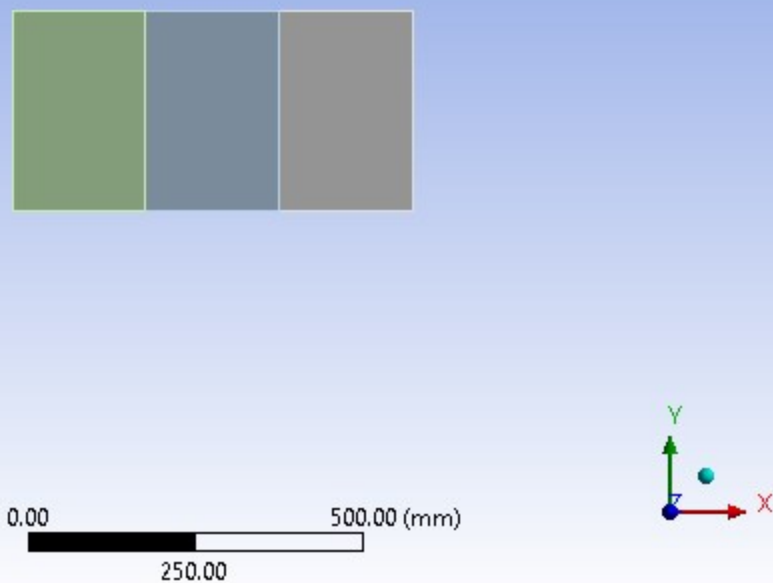


FIGURE 2
Model (A4) > Materials > Figure

Figure

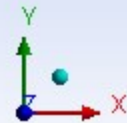
5/10/2024 11:33 PM

Ansys
2021 R2

- Aluminum Alloy
- Bronze, C51000
- Structural Steel



0.00 500.00 (mm)
250.00



Coordinate Systems

TABLE 5**Model (A4) > Coordinate Systems > Coordinate System**

Object Name	<i>Global Coordinate System</i>
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. mm
Origin Y	0. mm
Directional Vectors	
X Axis Data	[1. 0.]
Y Axis Data	[0. 1.]

Connections

TABLE 6
Model (A4) > Connections

Object Name	<i>Connections</i>
State	Fully Defined
Auto Detection	
Generate Automatic Connection On Refresh	Yes
Transparency	
Enabled	Yes

TABLE 7
Model (A4) > Connections > Contacts

Object Name	<i>Contacts</i>
State	Fully Defined
Definition	
Connection Type	Contact
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Auto Detection	
Tolerance Type	Slider
Tolerance Slider	0.
Tolerance Value	1.6771 mm
Use Range	No
Edge/Edge	Yes
Priority	Include All
Group By	Bodies
Search Across	Bodies
Statistics	
Connections	2
Active Connections	2

TABLE 8
Model (A4) > Connections > Contacts > Contact Regions

Object Name	Contact Region	Contact Region 2
State	Fully Defined	
Scope		
Scoping Method	Geometry Selection	
Contact	1 Edge	
Target	1 Edge	
Contact Bodies	2D	
Target Bodies	2D	
Shell Thickness Effect	No	
Protected	No	
Definition		
Type	Bonded	
Scope Mode	Automatic	
Behavior	Program Controlled	
Trim Contact	Program Controlled	
Trim Tolerance	1.6771 mm	

Suppressed	No
Display	
Element Normals	No
Advanced	
Formulation	Program Controlled
Small Sliding	Program Controlled
Detection Method	Program Controlled
Elastic Slip Tolerance	Program Controlled
Thermal Conductance	Program Controlled
Pinball Region	Program Controlled
Geometric Modification	
Contact Geometry Correction	None
Target Geometry Correction	None
Flip Contact Normals	No
Flip Target Normals	No

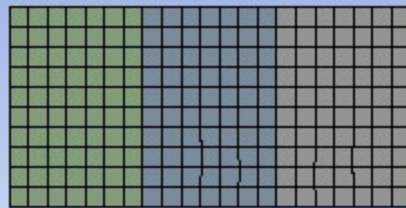
Mesh

TABLE 9
Model (A4) > Mesh

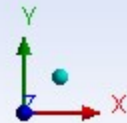
Object Name	<i>Mesh</i>
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	Default (30.619 mm)
Sizing	
Use Adaptive Sizing	No
Growth Rate	Default (1.2)
Mesh Defeaturing	Yes
Defeature Size	Default (0.15309 mm)
Capture Curvature	Yes
Curvature Min Size	Default (0.30619 mm)
Curvature Normal Angle	Default (30.0°)
Capture Proximity	No
Bounding Box Diagonal	670.82 mm
Average Surface Area	60000 mm ²
Minimum Edge Length	200.0 mm
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Quality	Default (0.050000)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None

Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	2
Growth Rate	1.2
Inflation Algorithm	Pre
View Advanced Options	No
Batch Connections	
Mesh Based Connection	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Use Sheet Thickness for Pinch	No
Pinch Tolerance	Default (0.27557 mm)
Generate Pinch on Refresh	No
Sheet Loop Removal	No
Statistics	
Nodes	735
Elements	210

FIGURE 3
Model (A4) > Mesh > Figure



0.00 250.00 500.00 (mm)



Steady-State Thermal (A5)

TABLE 10
Model (A4) > Analysis

Object Name	<i>Steady-State Thermal (A5)</i>
State	Solved
Definition	
Physics Type	Thermal
Analysis Type	Steady-State
Solver Target	Mechanical APDL
Options	
Generate Input Only	No

TABLE 11
Model (A4) > Steady-State Thermal (A5) > Initial Condition

Object Name	<i>Initial Temperature</i>
State	Fully Defined
Definition	
Initial Temperature	Uniform Temperature

Initial Temperature Value	22. °C
---------------------------	--------

TABLE 12
Model (A4) > Steady-State Thermal (A5) > Analysis Settings

Object Name	<i>Analysis Settings</i>
State	Fully Defined
Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Solver Pivot Checking	Program Controlled
Radiosity Controls	
Radiosity Solver	Program Controlled
Flux Convergence	1.e-004
Maximum Iteration	1000.
Solver Tolerance	1.e-007 W/mm ²
Over Relaxation	0.1
View Factor Method	Hidden
Number of Zones	200.
Nonlinear Controls	
Heat Convergence	Program Controlled
Temperature Convergence	Program Controlled
Line Search	Program Controlled
Advanced	
Contact Split (DMP)	Off
Output Controls	
Calculate Thermal Flux	Yes
Contact Data	Yes
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\90551\Desktop\5ler\18\2D_files\dp0\SYS\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	Yes
Solver Units	Active System
Solver Unit System	nmm

FIGURE 4
Model (A4) > Steady-State Thermal (A5) > Figure

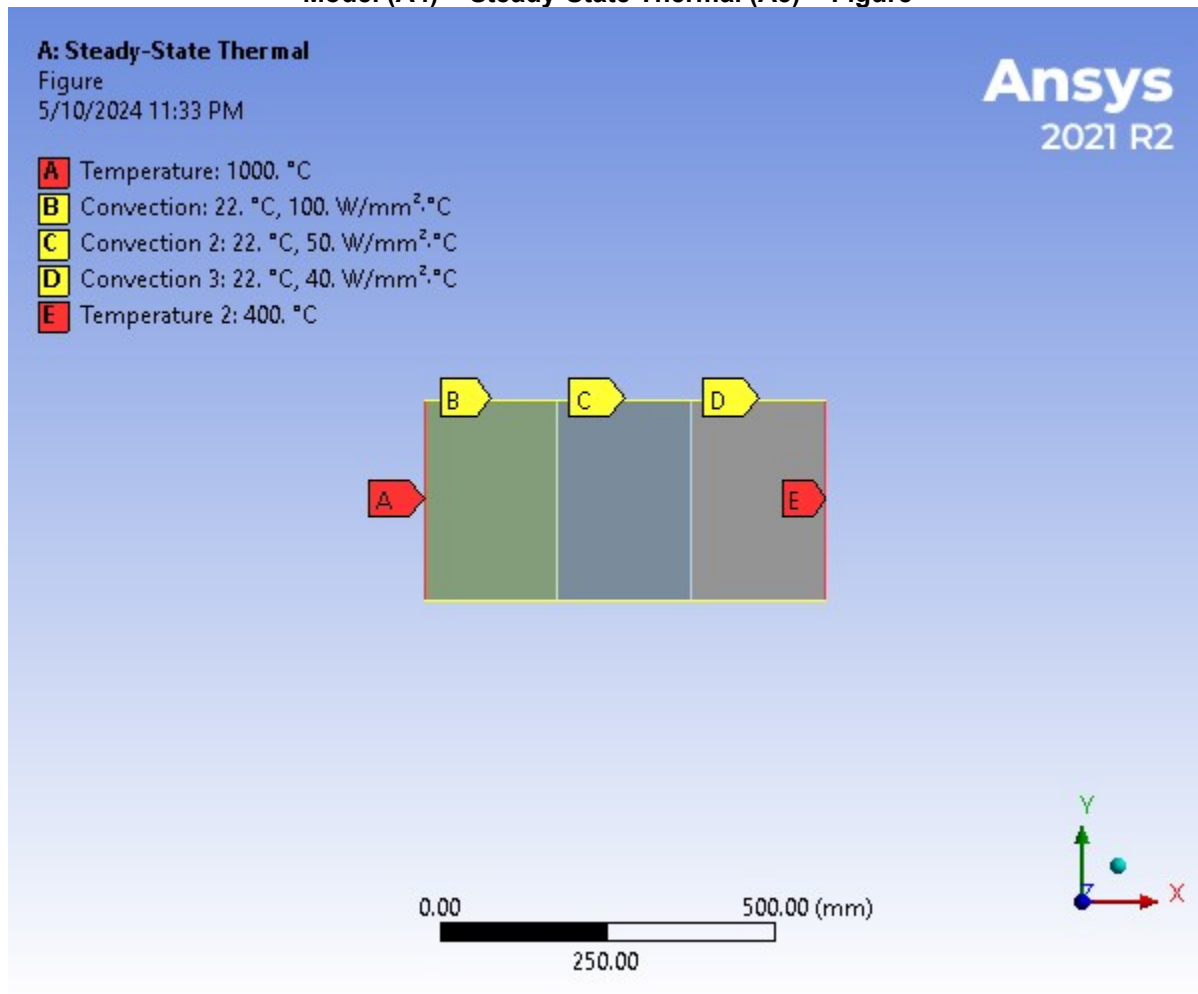


TABLE 13
Model (A4) > Steady-State Thermal (A5) > Loads

Object Name	Temperature	Convection	Convection 2	Convection 3	Temperature ₂
State	Fully Defined				
Scope					
Scoping Method	Geometry Selection				
Geometry	1 Edge	2 Edges			1 Edge
Definition					
Type	Temperature	Convection			Temperature
Magnitude	1000. °C (ramped)				400. °C (ramped)
Suppressed	No				
Film Coefficient		100. W/mm ² .°C (step applied)	50. W/mm ² .°C (step applied)	40. W/mm ² .°C (step applied)	
Ambient Temperature		22. °C (ramped)			

Convection Matrix		Program Controlled	
----------------------	--	--------------------	--

FIGURE 5
Model (A4) > Steady-State Thermal (A5) > Temperature

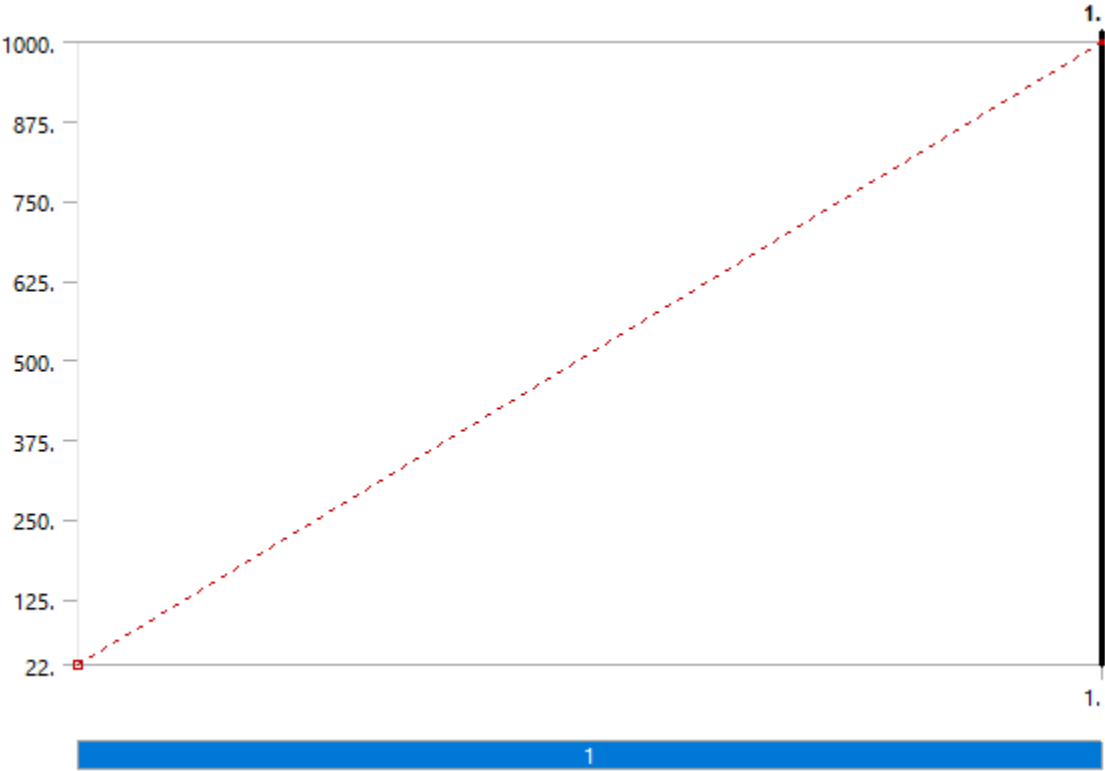


FIGURE 6
Model (A4) > Steady-State Thermal (A5) > Convection

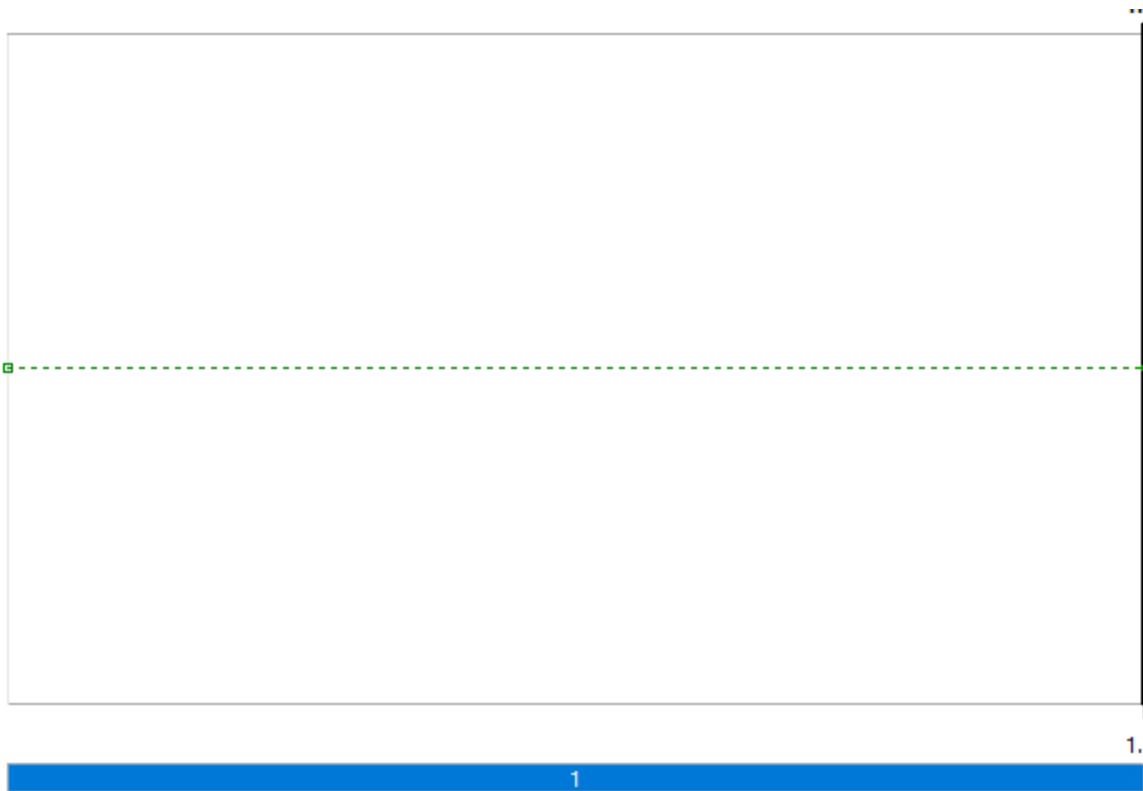


TABLE 14
Model (A4) > Steady-State Thermal (A5) > Convection

Steps	Time [s]	Convection Coefficient [W/mm ² .°C]	Temperature [°C]
1	0.	= 100.	= 22.
	1.	100.	22.

FIGURE 7
Model (A4) > Steady-State Thermal (A5) > Convection 2

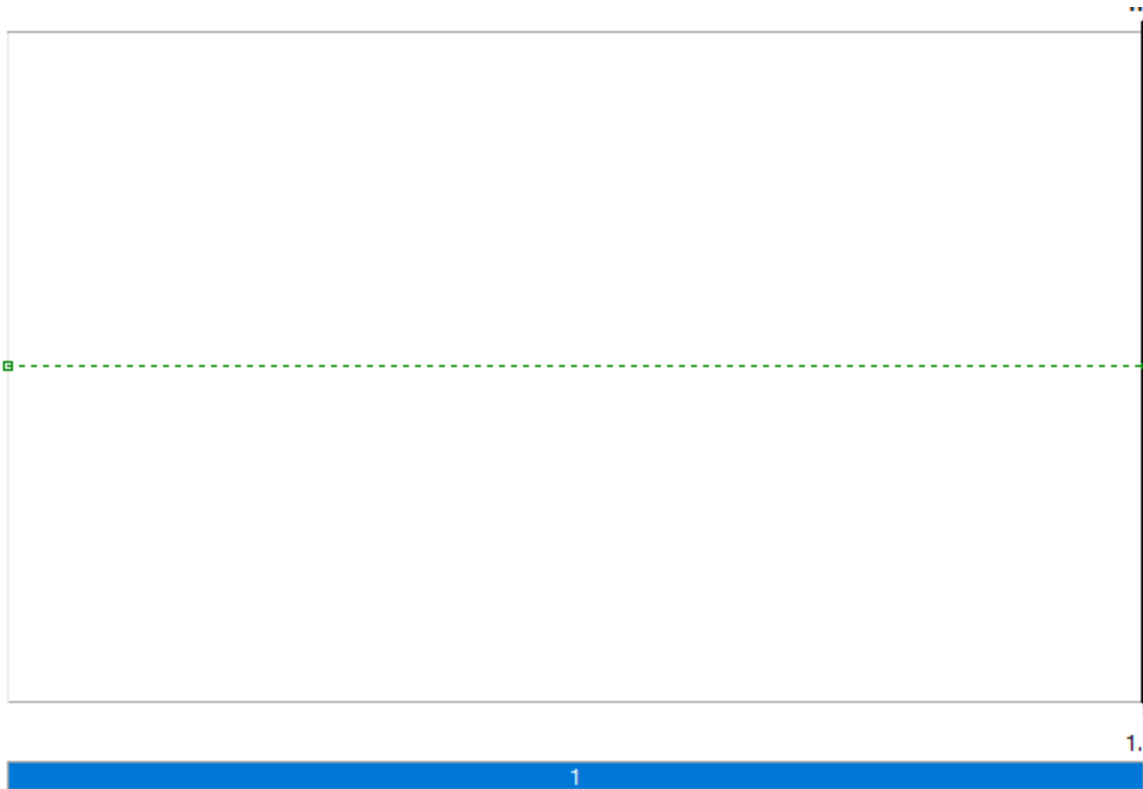


TABLE 15
Model (A4) > Steady-State Thermal (A5) > Convection 2

Steps	Time [s]	Convection Coefficient [W/mm ² .°C]	Temperature [°C]
1	0.	= 50.	= 22.
	1.	50.	22.

FIGURE 8
Model (A4) > Steady-State Thermal (A5) > Convection 3

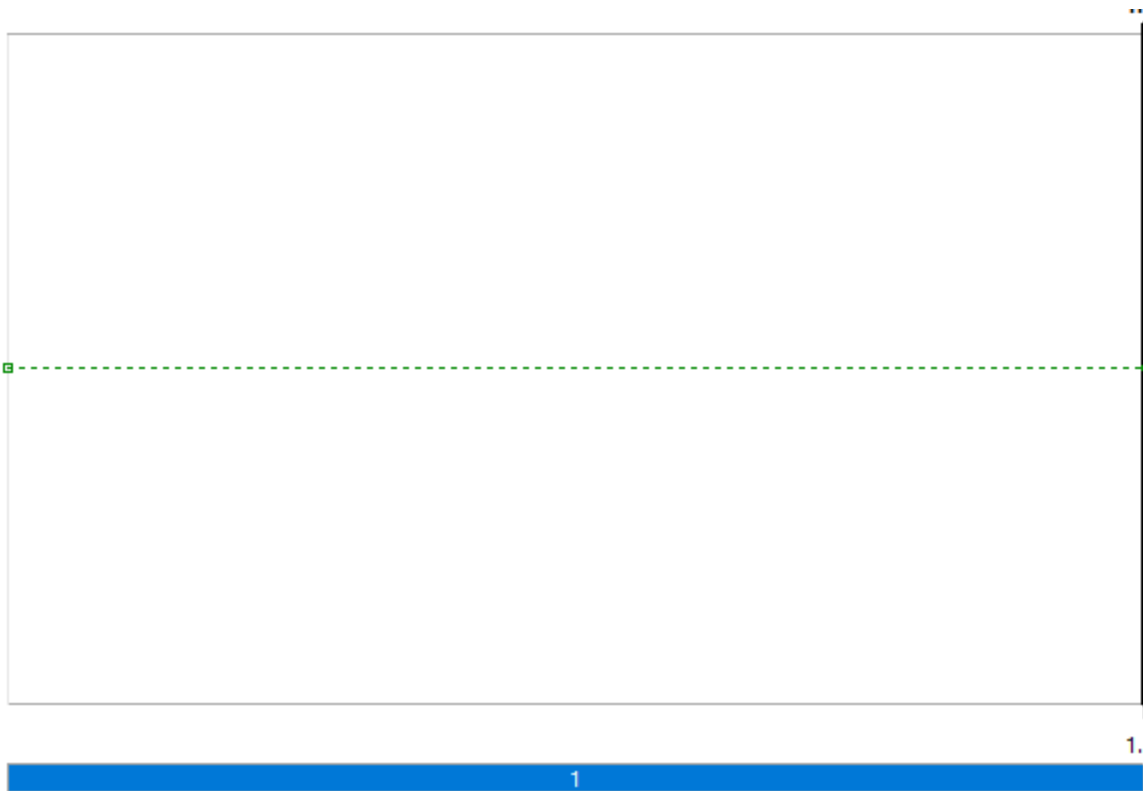
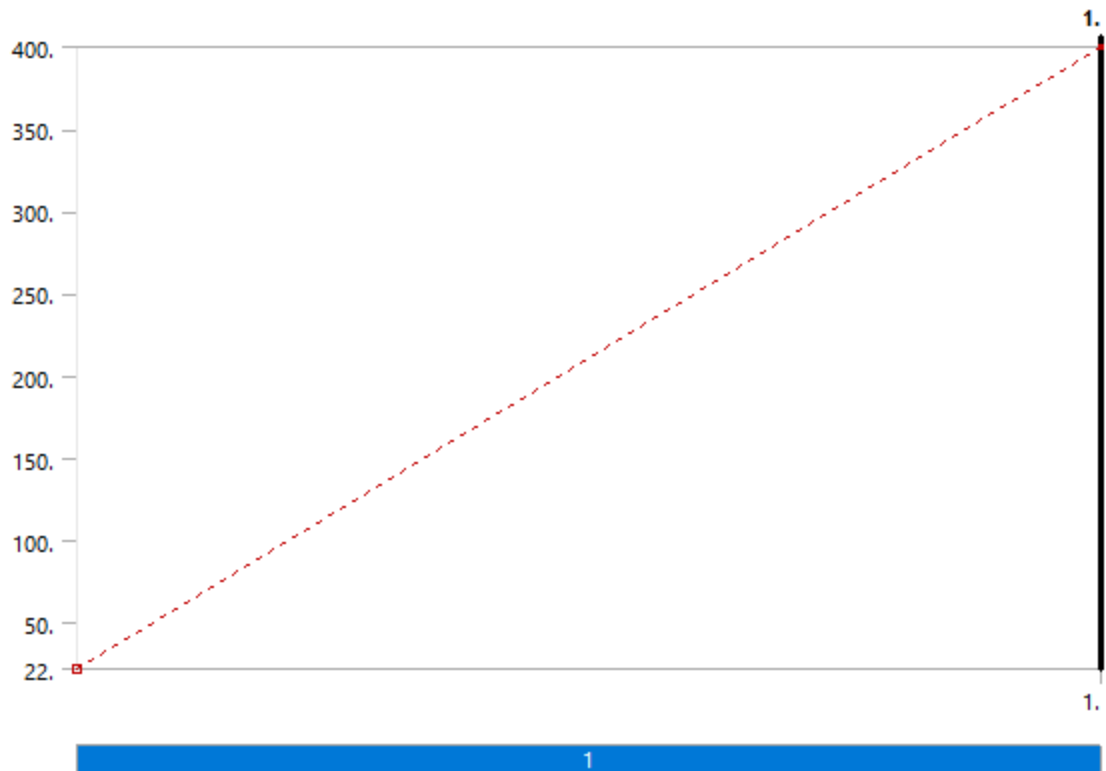


TABLE 16
Model (A4) > Steady-State Thermal (A5) > Convection 3

Steps	Time [s]	Convection Coefficient [W/mm ² ·°C]	Temperature [°C]
1	0.	= 40.	= 22.
	1.	40.	22.

FIGURE 9
Model (A4) > Steady-State Thermal (A5) > Temperature 2



Solution (A6)

TABLE 17
Model (A4) > Steady-State Thermal (A5) > Solution

Object Name	<i>Solution (A6)</i>
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	2. s
MAPDL Memory Used	86. MB
MAPDL Result File Size	448. KB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 18
Model (A4) > Steady-State Thermal (A5) > Solution (A6) > Solution Information

Object Name	<i>Solution Information</i>
State	Solved
Solution Information	
Solution Output	Solver Output
Update Interval	2.5 s

Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 19
Model (A4) > Steady-State Thermal (A5) > Solution (A6) > Results

Object Name	<i>Temperature</i>
State	Solved
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Definition	
Type	Temperature
By	Time
Display Time	Last
Calculate Time History	Yes
Identifier	
Suppressed	No
Results	
Minimum	-121.13 °C
Maximum	1000. °C
Average	192.74 °C
Minimum Occurs On	2D
Maximum Occurs On	2D
Information	
Time	1. s
Load Step	1
Substep	1
Iteration Number	1

FIGURE 10
Model (A4) > Steady-State Thermal (A5) > Solution (A6) > Temperature

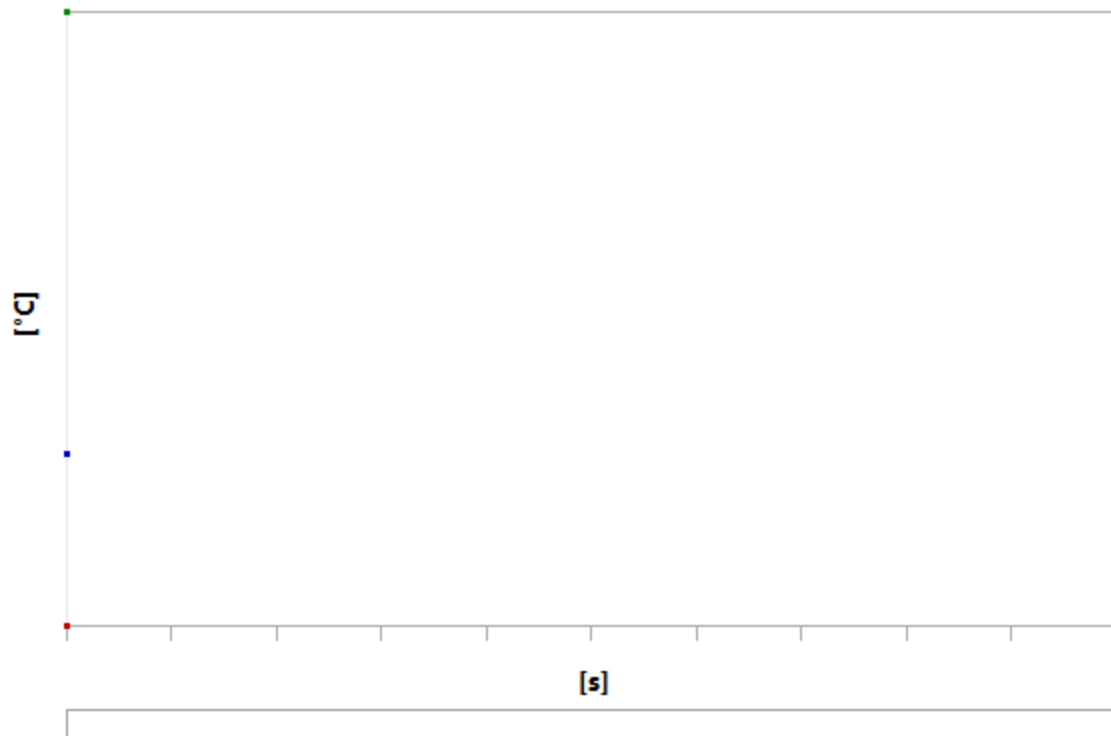
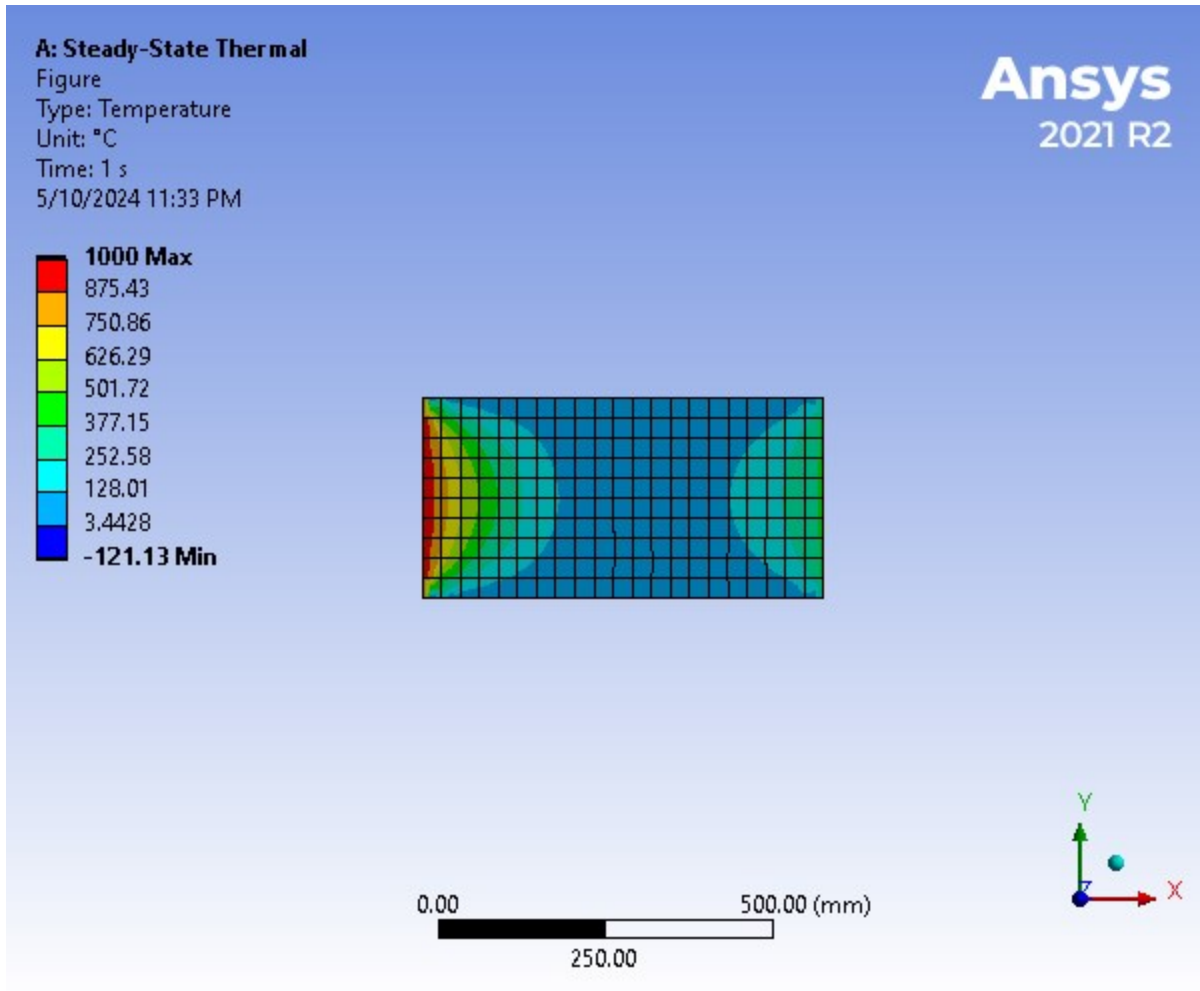


TABLE 20
Model (A4) > Steady-State Thermal (A5) > Solution (A6) > Temperature

Time [s]	Minimum [°C]	Maximum [°C]	Average [°C]
1.	-121.13	1000.	192.74

FIGURE 11
Model (A4) > Steady-State Thermal (A5) > Solution (A6) > Temperature > Figure



Material Data

Structural Steel

TABLE 21
Structural Steel > Constants

Density	7.85e-006 kg mm ⁻³
Coefficient of Thermal Expansion	1.2e-005 C ⁻¹
Specific Heat	4.34e+005 mJ kg ⁻¹ C ⁻¹
Thermal Conductivity	6.05e-002 W mm ⁻¹ C ⁻¹
Resistivity	1.7e-004 ohm mm

TABLE 22
Structural Steel > Color

Red	Green	Blue
132	139	179

TABLE 23
Structural Steel > Compressive Ultimate Strength

Compressive Ultimate Strength MPa
0

TABLE 24
Structural Steel > Compressive Yield Strength

Compressive Yield Strength MPa
250

TABLE 25
Structural Steel > Tensile Yield Strength

Tensile Yield Strength MPa
250

TABLE 26
Structural Steel > Tensile Ultimate Strength

Tensile Ultimate Strength MPa
460

TABLE 27
Structural Steel > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature C
22

TABLE 28
Structural Steel > S-N Curve

Alternating Stress MPa	Cycles	Mean Stress MPa
3999	10	0
2827	20	0
1896	50	0
1413	100	0
1069	200	0
441	2000	0
262	10000	0
214	20000	0
138	1.e+005	0
114	2.e+005	0
86.2	1.e+006	0

TABLE 29
Structural Steel > Strain-Life Parameters

Strength Coefficient MPa	Strength Exponent	Ductility Coefficient	Ductility Exponent	Cyclic Strength Coefficient MPa	Cyclic Strain Hardening Exponent
920	-0.106	0.213	-0.47	1000	0.2

TABLE 30
Structural Steel > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature C
2.e+005	0.3	1.6667e+005	76923	

TABLE 31
Structural Steel > Isotropic Relative Permeability

Relative Permeability
10000

Aluminum Alloy

TABLE 32
Aluminum Alloy > Constants

Density	2.77e-006 kg mm ⁻³
Coefficient of Thermal Expansion	2.3e-005 C ⁻¹
Specific Heat	8.75e+005 mJ kg ⁻¹ C ⁻¹

TABLE 33
Aluminum Alloy > Color

Red	Green	Blue
138	104	46

TABLE 34
Aluminum Alloy > Compressive Ultimate Strength

Compressive Ultimate Strength MPa
0

TABLE 35
Aluminum Alloy > Compressive Yield Strength

Compressive Yield Strength MPa
280

TABLE 36
Aluminum Alloy > Tensile Yield Strength

Tensile Yield Strength MPa
280

TABLE 37
Aluminum Alloy > Tensile Ultimate Strength

Tensile Ultimate Strength MPa
310

TABLE 38
Aluminum Alloy > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature C
22

TABLE 39
Aluminum Alloy > Isotropic Thermal Conductivity

Thermal Conductivity W mm ⁻¹ C ⁻¹	Temperature C
0.114	-100
0.144	0
0.165	100
0.175	200

TABLE 40
Aluminum Alloy > S-N Curve

Alternating Stress MPa	Cycles	R-Ratio
275.8	1700	-1
241.3	5000	-1
206.8	34000	-1
172.4	1.4e+005	-1
137.9	8.e+005	-1
117.2	2.4e+006	-1
89.63	5.5e+007	-1
82.74	1.e+008	-1
170.6	50000	-0.5
139.6	3.5e+005	-0.5
108.6	3.7e+006	-0.5
87.91	1.4e+007	-0.5
77.57	5.e+007	-0.5
72.39	1.e+008	-0.5
144.8	50000	0
120.7	1.9e+005	0
103.4	1.3e+006	0
93.08	4.4e+006	0
86.18	1.2e+007	0
72.39	1.e+008	0
74.12	3.e+005	0.5
70.67	1.5e+006	0.5
66.36	1.2e+007	0.5
62.05	1.e+008	0.5

TABLE 41
Aluminum Alloy > Isotropic Resistivity

Resistivity ohm mm	Temperature C
2.43e-005	0
2.67e-005	20
3.63e-005	100

TABLE 42
Aluminum Alloy > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature C
71000	0.33	69608	26692	

TABLE 43
Aluminum Alloy > Isotropic Relative Permeability

Relative Permeability
1

Bronze, C51000

TABLE 44
Bronze, C51000 > Constants

Density	8.715e-006 kg mm ⁻³
Tensile Yield Strength	505 MPa
Tensile Ultimate Strength	574.5 MPa
Coefficient of Thermal Expansion	1.675e-005 C ⁻¹
Thermal Conductivity	7.735e-002 W mm ⁻¹ C ⁻¹
Specific Heat	3.845e+005 mJ kg ⁻¹ C ⁻¹

TABLE 45
Bronze, C51000 > Opacity

Red	Green	Blue
204	102	0
Opacity		
1		
Metallic Finish		
1		

TABLE 46
Bronze, C51000 > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature C
1.077e+005	0.345	1.1581e+005	40037	23

TABLE 47
Bronze, C51000 > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature C
20